

Mohamed Abdellatif Etman

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PROFESSIONAL SUMMARY

Innovative AI Engineer with expertise in designing and deploying end-to-end solutions using Generative AI, multimodal analysis, and cloud technologies. Proven ability to build and scale complex AI systems leveraging state-of-the-art models (GPT-4o, Gemini, Llama 3), RAG pipelines, and vector databases to solve real-world business problems. Adept at full-stack development, API design, and MLOps with strong focus on creating robust, scalable, and efficient applications on Microsoft Azure. Successfully delivered multiple AI-powered solutions and extensive portfolio of mobile applications driving measurable improvements in operational efficiency and user experience.

EDUCATION

Diploma in AI, Data Science, and Cloud Computing January 2024 – December 2024
Digital Egypt Builders Initiative (DEBI), Ministry of Communication Hybrid, Egypt

- Intensive 12-month program covering Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, and MLOps.

Bachelor's Degree in Computer Science July 2019 – July 2023
Faculty of Computers and Information, Tanta University Tanta, Egypt

- Relevant Coursework: Algorithms, Data Structures, Database Systems, Software Engineering, Artificial Intelligence, Machine Learning.

WORK EXPERIENCE

AI Engineer October 2025 – Present
Integra Pro Solutions Cairo, Egypt

- Develop AI solutions integrated with Odoo ERP for sales forecasting, customer analytics, and business process automation.
- Build full-stack applications using React Native, Django, Odoo, and PHP with custom module development and API integration.
- Design AI-powered Odoo modules leveraging Generative AI and ML models for intelligent business workflows.

AI Engineer (Freelance) December 2024 – October 2025
Independent Contractor Remote

- Delivered end-to-end AI solutions using GPT-4o, Gemini, and RAG pipelines for tourism and enterprise clients.
- Built scalable backend APIs with FastAPI, Redis, and Azure Cognitive Services supporting multilingual NLP.
- Integrated WhatsApp Business API and Meta Graph API for omnichannel conversational AI experiences.

AI Developer – MCIT Innovation Project November 2024 – March 2025
Ministry of Communications and Information Technology (MCIT) Remote, Egypt

- Led team to develop multimodal AI system for employee performance evaluation using video, audio, and text analysis.
- Managed ML pipeline from data collection to Azure deployment, achieving 40% processing time reduction.
- Prototype recognized by MCIT leadership for innovation in government digital transformation.

GenAI Engineer (Internship) July 2024 – October 2024
National Bank of Egypt (NBE) Remote, Egypt

- Led team to develop AI-powered financial dashboard analyzer using Vision Transformers and NLP.
- Built desktop application with voice-enabled features deployed across banking terminals.
- Achieved 60% reduction in financial analyst review time.

Full-Stack Developer & Mobile App Team Lead August 2022 – October 2025
Camp Coding Tanta, Egypt

- Led development of cross-platform mobile applications using React Native with significant user adoption.
- Architected backend services using PHP and MySQL with optimized database performance.
- Managed full software development lifecycle using Agile methodologies and mentored junior developers.

CORE COMPETENCIES

Generative AI • Large Language Models • RAG Systems • MLOps • Azure Cloud • API Development • Computer Vision • Natural Language Processing • Full-Stack Development • FastAPI • Docker • Machine Learning • Deep Learning • Microservices • Vector Databases

KEY PROJECTS

Generative AI System for Performance Analytics (MCIT Government Project)

- **Overview:** Co-engineered a comprehensive AI solution for the Ministry of Communications automating employee performance reviews by synthesizing multimodal data from video, audio, and text sources.
- **Key Contributions:** Orchestrated end-to-end pipelines combining computer vision (OpenPose, MediaPipe) for gesture recognition, audio analysis (Whisper) for speech transcription, and NLP for sentiment analysis. Optimized report generation by implementing Llama 3 and Qwen-VL to translate behavioral data into structured feedback, processing extensive evaluation sessions.
- **Impact:** Substantially reduced evaluation time, improved consistency in performance assessments, and provided data-driven insights for HR decision-making.
- **Technologies:** Python, PyTorch, OpenPose, MediaPipe, Whisper, Llama 3, Gemini 2.0 Flash, Azure Cloud, FastAPI, Docker.

RAG-Powered Conversational AI Assistant (Travel Agency Client)

- **Overview:** Designed and deployed complete backend architecture for an intelligent tourism chatbot capable of handling complex multi-turn conversations across WhatsApp and Web platforms, serving extensive customer interactions.
- **Key Contributions:** Architected high-precision RAG system using ChromaDB achieving excellent retrieval accuracy from large document collections. Engineered asynchronous API gateway using FastAPI to handle numerous concurrent requests with Redis session persistence. Integrated Meta Graph API webhooks for real-time WhatsApp message processing.
- **Impact:** Achieved high customer satisfaction scores, substantially reduced response time, and successfully handled significant portion of customer inquiries autonomously.
- **Technologies:** Gemini, GPT-4o, RAG, ChromaDB, LangChain, Azure Cognitive Services, FastAPI, Redis, Docker, WhatsApp Business API.

Automated Financial Insight Generator (National Bank of Egypt)

- **Overview:** Developed production-grade desktop application utilizing Vision Transformers to extract and analyze financial data from chart images, enabling natural language queries via voice and text interfaces.
- **Key Contributions:** Deployed Vision Transformers (Donut) achieving high accuracy in extracting numerical data from complex financial charts. Built intelligent Q&A system combining FAISS vector search with GPT-NeoX-20B for contextual financial analysis. Developed user-friendly PyQt5 desktop application with Azure TTS integration for voice-enabled experience.
- **Impact:** Substantially reduced financial analyst review time, processed extensive chart collections, and enabled faster decision-making for banking operations.
- **Technologies:** Transformers, Donut, FAISS, GPT-NeoX-20B, FastAPI, Azure Cognitive Services, PyQt5, Python.

TECHNICAL SKILLS

Generative AI & LLMs: GPT-4o, Gemini, Claude, Llama 3, Qwen-VL, Retrieval-Augmented Generation (RAG), Vector Databases (FAISS, ChromaDB, Pinecone), LangChain, Prompt Engineering

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Transformers, BERT, CNNs, RNNs, LSTMs, Model Fine-tuning, Transfer Learning

Natural Language Processing: spaCy, NLTK, Hugging Face, Text Classification, Sentiment Analysis, Named Entity Recognition, Text Generation

Computer Vision & Multimodal AI: OpenCV, OpenPose, MediaPipe, YOLO, Vision Transformers, Image Classification, Object Detection

Audio Processing: Whisper, librosa, Speech-to-Text, Audio Feature Extraction, Voice Analysis

Programming Languages: Python (Advanced), JavaScript, TypeScript, PHP, SQL, Bash

Backend Development: FastAPI, Django, Flask, Node.js, RESTful APIs, WebSocket, Async Programming, Redis, Celery

ERP & Business Automation: Odoo API Integration (XML-RPC, JSON-RPC), Custom Module Development, ORM, Business Logic Implementation

Databases: MongoDB, MySQL, PostgreSQL, Vector Databases, Database Optimization, Query Performance Tuning

Frontend & Mobile Development: React Native, React.js, Streamlit, PyQt5, Responsive Design

Cloud Platforms & Services: Microsoft Azure (Cognitive Services, App Service, Functions, Blob Storage), Azure OpenAI, Azure ML

MLOps & DevOps: Docker, Docker Compose, Git, GitHub, CI/CD Pipelines, MLflow, Hugging Face Hub, Model Deployment, Monitoring

Tools & Technologies: Postman, Jupyter Notebook, VS Code, Linux, Agile Methodologies

CERTIFICATIONS

- **Microsoft Azure AI-102:** AI Engineer Associate – [View Certificate](#)
- **Stanford Machine Learning Specialization** (Coursera) – Andrew Ng’s comprehensive ML course covering supervised, unsupervised learning, and neural networks.
- **Deep Learning Specialization** (Coursera) – Advanced deep learning including CNNs, RNNs, LSTMs, and Transformers.
- **Generative AI with Large Language Models** (Coursera) – Specialized training in LLM architecture, fine-tuning, and deployment.
- **MLOps Tools & Best Practices** (DEBI) – Practical experience with MLflow, Hugging Face, Docker, and CI/CD for ML systems.

LANGUAGES

- **Arabic:** Native Speaker
- **English:** Professional Working Proficiency (Written and Spoken)

ADDITIONAL INFORMATION

- **Military Status:** Exempted
- **Availability:** Immediate
- **Work Authorization:** Egyptian Citizen, available for remote and on-site opportunities